

SCI225 · PATHOPHYSIOLOGY · WEEK 12 · NIGHTINGALE COLLEGE

The Reproductive Systems

Female and male reproductive physiology and its disorders — the menstrual cycle, PCOS and endometriosis, testicular and prostate disease, and the hormonal logic that ties them together.

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Module 12

Female · Male

8 topics + cheat sheet

Female Reproductive System

SECTION 1

The cycle, its hormonal control, and the disorders of ovulation, structure, infection, and menopause.

The cycle is a hormonal relay: the **hypothalamus** → **pituitary** → **ovary** → **uterus**. Get the sequence and you can predict almost every disorder.

Hormone (source)	Job
GnRH (hypothalamus)	Triggers the pituitary to release FSH & LH; rises at puberty to start reproductive function
FSH (pituitary)	Stimulates ovarian follicles to mature
LH (pituitary)	The LH surge triggers ovulation
Estrogen (follicle)	Builds up (proliferates) the endometrium
Progesterone (corpus luteum)	After ovulation, maintains/stabilizes the endometrium for possible pregnancy

What happens across the cycle

- FSH matures a follicle → estrogen rises → endometrium thickens.
- The **LH surge** → **ovulation** (egg released).
- The ruptured follicle becomes the **corpus luteum**, which makes **progesterone** to hold the lining.
- No pregnancy → corpus luteum degenerates → **progesterone falls** → the lining loses support and sheds = **menstruation**.
- If fertilization occurs, it usually happens in the **fallopian tube (ampulla)**.

Key idea: progesterone is the "stay" hormone. Its withdrawal (when the corpus luteum dies) removes endometrial support → the lining breaks down and sheds. Anything that prevents ovulation prevents corpus luteum formation → no progesterone → abnormal, unopposed estrogen effects (Topic 2).

MEMORY HOOK

FSH = Follicle Stimulating (grows the egg). **LH = "Launches" ovulation** (the surge). **Corpus luteum = progesterone = "pro-gestation"** (holds the lining for pregnancy).

PRACTICE

Q1. The corpus luteum degenerates and progesterone falls. Which mechanism leads to menstruation?

- A. Loss of endometrial support causing tissue breakdown
- B. Rising estrogen thickening the lining
- C. An LH surge triggering ovulation
- D. Increased progesterone stabilizing the lining

A is correct. When progesterone is withdrawn, the endometrium loses its hormonal support, breaks down, and sheds as menstrual flow.

PRACTICE

Q2. Which structure produces progesterone after ovulation, and where does fertilization most often occur?

- A. Corpus luteum; fallopian tube (ampulla)
- B. Ovarian follicle; uterus
- C. Endometrium; cervix
- D. Pituitary; ovary

A is correct. The corpus luteum (the ruptured follicle) secretes progesterone; fertilization typically occurs in the ampulla of the fallopian tube.

When ovulation or hormone signaling goes wrong, the cycle becomes irregular.

Disorder	Mechanism
PCOS	Anovulation + insulin resistance: high insulin raises ovarian androgens and disrupts the LH surge → eggs don't release
Primary dysmenorrhea	High prostaglandins → uterine cramping + reduced uterine blood flow → pain
Amenorrhea	Absent periods — from anovulation, hormonal, or structural causes
Abnormal uterine bleeding	Anovulation → no progesterone → estrogen keeps building the lining → it overgrows then sheds irregularly (skipped cycles then heavy bleeding)

PCOS in depth (Olivia's case)

Olivia (28) has 1 year of infertility, only 4–5 cycles/year, BMI 35, acne, **elevated insulin and testosterone**, and ovaries with many small peripheral follicles. The chain: **insulin resistance** → **high insulin** → **excess ovarian androgens** → **disrupted LH surge** → **chronic anovulation** → infertility, irregular bleeding, acne, and hirsutism.

Key idea — why PCOS bleeding is irregular: without ovulation there's **no corpus luteum and no progesterone**, but estrogen keeps stimulating the endometrium. The lining overgrows (unopposed estrogen) and eventually sheds erratically — skipped cycles followed by heavy bleeding. This is also why PCOS raises **endometrial hyperplasia/cancer** risk (Topic 3) and **type 2 diabetes** risk.

Pearl: the "monitor for" answer in PCOS is **endometrial cancer and type 2 diabetes** (from unopposed estrogen and insulin resistance, respectively). Dysmenorrhea's "mediator" answer is always **prostaglandins**.

PRACTICE

Q1. Which pathophysiologic mechanism best explains chronic anovulation in PCOS?

- A. Excess insulin raises androgens and disrupts the LH surge
- B. Excess insulin promotes normal follicular maturation
- C. Low androgens prevent follicle growth
- D. High progesterone blocks ovulation

A is correct. Insulin resistance drives high insulin, which raises ovarian androgens and disrupts the LH surge needed to ovulate — so follicles don't release (anovulation).

PRACTICE

Q2. A patient reports skipped cycles followed by heavy uterine bleeding. Which process is the cause?

- A. Anovulation with endometrial overgrowth (unopposed estrogen)
- B. Estrogen loss thinning the lining
- C. Excess progesterone stabilizing the lining
- D. High prolactin maintaining the cycle

A is correct. Without ovulation there's no progesterone, so estrogen keeps thickening the endometrium until it overgrows and sheds irregularly — skipped cycles then heavy bleeding.

PRACTICE

Q3. Which mediator causes the uterine cramping and reduced blood flow of primary dysmenorrhea?

- A. Prostaglandins
- B. Progesterone
- C. FSH
- D. Insulin

A is correct. High prostaglandins cause strong uterine contractions and vasoconstriction (reduced blood flow) → the cramping pain of primary dysmenorrhea.

These involve abnormal tissue — ectopic, overgrown, or benign tumors.

Disorder	What it is	Consequence
Endometriosis	Ectopic endometrial tissue outside the uterus that still responds to cyclic hormones	Bleeds cyclically → inflammation, pelvic adhesions , pain, infertility
Uterine fibroids (leiomyoma)	Benign smooth-muscle tumors of the uterus	Heavy bleeding, bulk/pressure symptoms
Endometrial hyperplasia/cancer	Overgrowth from prolonged estrogen without progesterone	Abnormal bleeding; can progress to cancer
Androgen insensitivity syndrome	Genetically male, body can't respond to androgens; no uterus develops	No menstruation (no endometrium to shed)

Key idea — unopposed estrogen: estrogen *builds* the endometrium; progesterone *controls* it. **Estrogen for a long time without progesterone** (PCOS, anovulation, estrogen-only therapy) → the lining overgrows = **endometrial hyperplasia**, which can become endometrial cancer. This is why amenorrhea from anovulation still needs follow-up — the lining is unprotected.

Pearl — endometriosis vs fibroids: **endometriosis** = endometrial tissue in the *wrong place* (responds to hormones, bleeds, causes adhesions). **Fibroids (leiomyomas)** = benign *smooth-muscle* growths of the uterus itself.

PRACTICE

Q1. Which ectopic tissue responds to menstrual hormones, bleeds cyclically, and causes pelvic adhesions?

- A. Endometrial tissue outside the uterus (endometriosis)
- B. Smooth-muscle tissue (fibroid)
- C. Ovarian follicular tissue

D. Cervical epithelium

A is correct. Endometriosis is ectopic endometrial tissue that responds to the cycle, bleeds, and triggers inflammation and adhesions.

PRACTICE

Q2. A patient has prolonged estrogen exposure without progesterone. Which outcome occurs?

- A. Endometrial overgrowth (hyperplasia) with cancer risk
- B. Endometrial thinning
- C. Immediate menopause
- D. Suppressed estrogen receptors

A is correct. Unopposed estrogen keeps stimulating endometrial growth without progesterone to control it → hyperplasia, which can progress to endometrial cancer.

PRACTICE

Q3. Which missing structure prevents menstruation in androgen insensitivity syndrome?

- A. The uterus (no endometrium to shed)
- B. The ovaries
- C. The pituitary
- D. The cervix only

A is correct. Without a uterus there is no endometrial lining to build and shed, so menstruation cannot occur.

The vagina's normal acidity protects it; lose that protection and infection follows. Pelvic infection and menopause each change the tract.

Topic	Mechanism
Vaginitis	Disrupted normal flora / raised vaginal pH lets microorganisms overgrow
Douching	Washes out protective flora → ↑ pH → infection risk
Postmenopausal vaginitis	Low estrogen thins/atrophies the vaginal lining and raises pH → infection risk
PID → ectopic pregnancy	Infection scars the fallopian tubes ; scarring blocks/slows egg transport → implantation in the tube (ectopic)
Menopause	Ovarian estrogen declines → cycles stop; atrophy, vasomotor symptoms, bone loss

Key idea — PID and ectopic pregnancy: pelvic inflammatory disease scars and narrows the fallopian tubes. A fertilized egg can't move normally to the uterus, so it implants in the **tube** — an ectopic pregnancy. The mechanism is **fallopian tube scarring**.

Pearl: estrogen keeps the vagina acidic and the lining thick. **Less estrogen (menopause) or washed-out flora (douching) → higher pH → more infections.** The protective factor is the **acidic environment**.

PRACTICE

Q1. A patient with a history of PID is at increased ectopic pregnancy risk. Which change explains it?

- A. Fallopian tube scarring blocking normal egg transport
- B. Thickened endometrium
- C. Increased ovarian reserve

D. Elevated progesterone

A is correct. PID scars the fallopian tubes; the fertilized egg gets trapped and implants in the tube rather than the uterus.

PRACTICE

Q2. After menopause, why is vaginal infection risk increased?

- A. Decreased estrogen thins the lining and raises vaginal pH
- B. Increased estrogen thickens the lining
- C. Higher progesterone
- D. Increased glycogen and acidity

A is correct. Low estrogen causes atrophy and a higher (less acidic) pH, weakening the vagina's natural defense against microorganisms.

SECTION 2

Male Reproductive System

Erection and sperm physiology, the scrotal emergencies, prostate disease, and erectile dysfunction.

Three jobs: make **testosterone**, make and mature **sperm**, and achieve **erection** for delivery.

Structure / process	Role
Testes (Leydig cells)	Produce testosterone ; sperm form in the seminiferous tubules
Epididymis	Sperm mature and gain motility here
Sperm flagellum (tail)	Provides motility ; tail dysfunction → impaired sperm movement
Erection	A vascular event — arterial inflow fills the erectile tissue while venous outflow is restricted

Erection physiology

Erection is fundamentally about **blood filling the erectile (cavernous) tissue**: parasympathetic signals dilate arteries, blood floods the sinusoids, and the swelling compresses the veins to trap it. Anything that limits inflow (vascular disease) limits erection (Topic 8).

Aging male

Testosterone gradually **declines** with age → reduced libido, less muscle mass, slower sperm production, and a higher rate of erectile and prostate issues.

Key idea: erection = vascular filling; sperm motility = the flagellum + epididymal maturation. Separating these two helps you place each disorder: ED is usually a *blood-flow* problem; infertility from poor motility is a *sperm-structure/maturation* problem.

PRACTICE

Q1. A patient has impaired sperm motility. Dysfunction of which structure best explains this?

- A. The sperm flagellum (tail)
- B. The prostate gland
- C. The seminal vesicle duct
- D. The bladder neck

A is correct. The flagellum (tail) propels the sperm; if it's defective, motility drops. (Sperm also gain motility maturing in the epididymis.)

PRACTICE

Q2. Which physiologic change occurs during erection?

- A. Increased arterial blood flow fills the erectile tissue
- B. Venous outflow increases to drain the tissue
- C. Arterial constriction reduces inflow
- D. Smooth muscle of the corpora contracts to empty blood

A is correct. Erection is a vascular event: arterial inflow fills the cavernous tissue while venous outflow is compressed to maintain it.

The scrotal differential is high-yield — especially distinguishing the **surgical emergency** (torsion) from infections.

Disorder	Mechanism / clues
Testicular torsion	The spermatic cord twists → cuts off blood supply → ischemia . Sudden, severe pain (often during activity), nausea, testicle riding high . Surgical emergency .
Epididymitis	Often from an STI ; gradual scrotal pain/swelling + burning with urination (dysuria)
Orchitis	Testicular inflammation, classically after a viral illness (mumps) → fever + painful swelling
Varicocele	Dilated scrotal veins → pooled blood raises testicular temperature → reduced sperm quality
Cryptorchidism (undescended testis)	Testis fails to descend → later ↑ risk of infertility and testicular cancer
Gynecomastia	Male breast tissue from an estrogen-to-testosterone imbalance (relative estrogen excess)

Key idea — torsion vs epididymitis: **torsion** = *sudden*, severe pain, high-riding testicle, cord twist → ischemia — minutes matter (untwist surgically to save the testicle). **Epididymitis** = *gradual* pain with an STI and **burning urination**. Timing + dysuria separate them.

Pearl: **varicocele** raises scrotal temperature (warm blood pooling), and sperm production needs a **cooler** environment — that's why it lowers sperm quality. **Cryptorchidism's** exam answer is increased **cancer/infertility** risk.

PRACTICE

Q1. A 17-year-old has sudden severe unilateral scrotal pain during sports, nausea, and a testicle positioned higher than the other. Which process explains this?

- A. Spermatic cord twisting cutting off blood supply (ischemia)
- B. Gradual STI-related inflammation
- C. Dilated scrotal veins raising temperature
- D. Post-viral testicular inflammation

A is correct. Testicular torsion: the cord twists, blood supply is cut off, and the testicle is ischemic and high-riding — a surgical emergency.

PRACTICE

Q2. A 25-year-old has gradual unilateral scrotal pain, swelling, and burning with urination after unprotected sex. Which finding supports epididymitis?

- A. Dysuria (burning urination) with gradual onset and recent STI exposure
- B. Sudden onset during exertion with a high-riding testis
- C. Bilateral painless masses
- D. Absent cremasteric reflex with instant pain

A is correct. Epididymitis is usually an STI-related, gradual process with dysuria — unlike the sudden, ischemic onset of torsion.

PRACTICE

Q3. Which vascular change reduces sperm quality in varicocele?

- A. Dilated veins pool blood and raise testicular temperature
- B. Arterial blockage cools the testis
- C. Reduced venous pressure lowers temperature
- D. Increased lymphatic drainage

A is correct. Pooled venous blood warms the testis, and sperm production needs cooler-than-core temperature — so sperm quality drops.



The prostate wraps around the urethra, so prostate enlargement squeezes the urinary stream. Both BPH and prostate cancer can obstruct.

	BPH (benign prostatic hyperplasia)	Prostate cancer
What	Benign enlargement (very common with aging)	Malignant growth, often peripheral
Mechanism of urinary symptoms	Enlarged prostate compresses the urethra	Tumor growth compresses/ obstructs the urethra (often later)
Symptoms	Hesitancy, weak stream, incomplete emptying, nocturia	Similar obstructive symptoms; may be asymptomatic early; can metastasize (bone)

Key idea: the urinary symptoms of both BPH and prostate cancer come from the **same mechanism — the enlarged prostate compressing the urethra** → hesitancy, weak stream, incomplete emptying, nocturia. BPH is benign and extremely common with age; cancer must be ruled out (screening, exam, PSA).

PRACTICE

Q1. An older adult reports urinary hesitancy, a weak stream, and incomplete bladder emptying. What is the cause?

- A. An enlarged prostate compressing the urethra
- B. Bladder muscle hypertrophy increasing flow
- C. Urethral widening
- D. Decreased prostate size

A is correct. Prostate enlargement (most often BPH) compresses the urethra, producing hesitancy, weak stream, and incomplete emptying.

PRACTICE

Q2. A 72-year-old has a slow stream, hesitancy, and nocturia. Which mechanism explains how prostate cancer can cause these?

- A. Tumor growth compresses and obstructs the urethra
- B. The tumor widens the urethra
- C. The bladder shrinks the prostate
- D. Increased urine production only

A is correct. Like BPH, prostate cancer can compress/obstruct the urethra, producing obstructive urinary symptoms.



Because erection is a **vascular** event, most erectile dysfunction is a blood-flow problem.

Condition	Mechanism
Vascular ED	Reduced arterial inflow can't fill the erectile tissue (atherosclerosis, diabetes, hypertension)
Priapism	Prolonged, unwanted erection → trapped blood becomes deoxygenated → ischemia damages erectile tissue → later ED

Key idea: vascular ED = not enough inflow to fill the tissue (a "pipe" problem — often the first sign of systemic vascular disease/diabetes). **Priapism** is the opposite extreme: blood gets *stuck*, goes ischemic, and the resulting tissue damage causes later erectile dysfunction — which is why prolonged priapism is an emergency.

PRACTICE

Q1. Which change limits filling of the erectile tissue in vascular erectile dysfunction?

- A. Reduced arterial inflow to the erectile tissue
- B. Excess arterial inflow
- C. Increased venous trapping of blood
- D. Overactive parasympathetic dilation

A is correct. Vascular ED results from inadequate arterial inflow, so the cavernous tissue can't fill and stiffen.

PRACTICE

Q2. Why can prolonged priapism lead to erectile dysfunction?

- A. Trapped blood becomes ischemic and damages the erectile tissue

- B. Excess oxygen strengthens the tissue
- C. Increased arterial inflow heals the tissue
- D. Rapid venous drainage protects the tissue

A is correct. In prolonged priapism blood stays trapped, loses oxygen, and the resulting ischemia injures the erectile tissue — leading to later ED.

□ EXAM

Reproductive Cheat Sheet

The highest-yield patterns — review the night before.

Female — don't miss

- **Hormones:** GnRH → FSH/LH; **FSH grows follicle, LH surge = ovulation**; corpus luteum = **progesterone** (holds lining). Progesterone withdrawal → menstruation. Fertilization = fallopian tube (ampulla).
- **PCOS:** insulin resistance → ↑ androgens → disrupted LH surge → **anovulation**; irregular/heavy bleeding; monitor for **endometrial cancer & type 2 diabetes**.
- **Dysmenorrhea mediator = prostaglandins.**
- **Unopposed estrogen** (no progesterone) → endometrial **hyperplasia** → **cancer**.
- **Endometriosis** = ectopic endometrial tissue → cyclic bleeding, adhesions. **Fibroid (leiomyoma)** = benign smooth muscle.
- **PID** → fallopian tube scarring → **ectopic pregnancy**.
- **Vaginitis:** ↑ pH (douching, low estrogen after menopause) lets microbes overgrow. Estrogen keeps it acidic/protective.
- **Androgen insensitivity:** no uterus → no menstruation.

Male — don't miss

- **Erection = vascular filling** of erectile tissue; **vascular ED** = ↓ arterial inflow. **Priapism** → ischemia → later ED.

- **Torsion** = cord twist → ischemia; **sudden** severe pain, high-riding testis — **surgical emergency**.
- **Epididymitis** = STI, **gradual** pain + **dysuria**. **Orchitis** = after viral illness (mumps).
- **Varicocele** = dilated veins → ↑ temperature → ↓ sperm quality.
- **Cryptorchidism** (undescended testis) → ↑ infertility & testicular cancer risk.
- **Sperm motility** = **flagellum** (+ epididymal maturation). **Gynecomastia** = estrogen/testosterone imbalance.
- **BPH & prostate cancer**: enlarged prostate **compresses the urethra** → hesitancy, weak stream, incomplete emptying, nocturia.

□ TERMS

Glossary

Quick definitions of the Week 12 vocabulary.

GnRH / FSH / LH

Hypothalamic and pituitary hormones that drive the ovarian cycle; the LH surge triggers ovulation.

Corpus luteum

The ruptured follicle after ovulation; secretes progesterone to maintain the endometrium.

Anovulation

Failure to ovulate; no corpus luteum → no progesterone → irregular bleeding (PCOS).

Unopposed estrogen

Estrogen stimulating the endometrium without progesterone → hyperplasia/cancer risk.

PCOS

Polycystic ovary syndrome — insulin resistance, excess androgens, anovulation, infertility.

Dysmenorrhea

Painful menstruation; primary type driven by prostaglandins.

Endometriosis

Endometrial tissue outside the uterus that bleeds cyclically and causes adhesions.

Leiomyoma (fibroid)

Benign uterine smooth-muscle tumor.

PID

Pelvic inflammatory disease; scars fallopian tubes → ectopic pregnancy risk.

Ectopic pregnancy

Implantation outside the uterus, usually in a scarred fallopian tube.

Testicular torsion

Twisting of the spermatic cord cutting off blood supply — a surgical emergency.

Epididymitis / Orchitis

Inflammation of the epididymis (often STI) / testis (often post-viral, mumps).

Varicocele

Dilated scrotal veins that raise temperature and reduce sperm quality.

Cryptorchidism

Undescended testis; raises infertility and testicular cancer risk.

BPH

Benign prostatic hyperplasia; enlarged prostate compresses the urethra → obstructive urinary symptoms.

Priapism

Prolonged erection → ischemia → later erectile dysfunction.